Evaluating the Pollutant Removal Efficiency of a New Retrofittable Sustainable Drainage Material

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Abstract. Urban transport runoff is a significant contributor to water pollution. It contains a mixture of pollutants which originate from sources including fuel combustion, fuel losses and oil leakages from hydraulic systems. This research sought to evaluate a new sustainable drainage material, known as BlueLay, for its pollutant removal efficiency. Initially, the hydraulic performance of the BlueLay was tested in terms of infiltration volume, absorption capacity and runoff compared with asphalt slabs of the same area. Building on these findings, a laboratory-scale pollutant test rig was constructed to evaluate the pollutant removal efficiency of BlueLay. The material was subjected to typical concentrations of Total Suspended Solids (TSS) (i.e., 75–774 mg/L) and a subset of heavy metals – Cadmium (Cd), Copper (Cu), Zinc (Zn) and Lead (Pb) found in road runoff under scenarios that replicated first flush and saturated overland flow events. The findings revealed that the range of TSS removal by BlueLay was >94% for all scenarios, and > 97% removal was achieved for saturated BlueLay. BlueLay in its dry state treated Pb (<145 µg/L), Cu (<110 µg/L), Zn (<315 µg/L) and Cd (<5 µg/L) to values well within the legislative (S.I. 272/2009) limits. For the highest applied concentrations of Cu (170 µg/L), Zn (600 µg/L) and Cd (10 µg/L), dry BlueLay exceeded the legislative limits by 1.4 µg/L, 9.2 µg/L and 0.4 µg/L respectively. In comparision, the exceedance from asphalt for the same metal concentrations, was approximately 100x, 50x and 20x larger. Future work includes testing pollutant removal efficiencies for a subset of Polyaromatic hydrocarbons.

Keywords: Water Quality · Road Runoff · Sustainable Drainage

1 Introduction

An increase in urban development globally has resulted in decreased infiltration and increased surface runoff [1, 2]. This is a major cause of water pollution in rapidly growing cities [3] as pollutants present on impervious urban surfaces get washed into waterways along with the excess water [4]. In the UK, urban transport runoff contributes to nearly 13% of all water pollution [5]. Conventional stormwater drains are designed to mitigate flood risk, but do not treat pollutants present in surface runoff before conveying it into waterways [3].

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Fuel combustion, oil spills, road construction materials [6], and vehicle component wear [7, 8] are some sources that can introduce pollutants onto road surfaces [9]. The composition of pollutants depends on factors such as Average Annual Daily Traffic (AADT) [10], land use, wind, storm events and dry periods [11]. Pollutants in road runoff include Poly Aromatic Hydrocarbons (PAHs) [12], vehicle oils [13], Volatile Organic Carbons (VOCs), heavy metals and Total Suspended Solids (TSS) among others. These can be water-soluble, or sediment bound. TSS and heavy metals [14, 15] are highly prevalent in road runoff [16] and designated as priority/specific pollutants [17, 18] and pose a major ecological risk [19, 20] to freshwater ecosystems. Road deposited sediment (RDS) accumulated in dry weather is mobilized during rainfall events and can act as a carrier [21, 22] for pollutants and as a source of TSS in surface runoff [23]. Pollutants are regulated to comply with various legislative standards globally. Standards followed in Ireland are set by the European Union Water Framework Directive (WFD) [17] and the S.I. 272/2009 [18].

High intensity stormflow conditions in Ireland are predicted to increase by up to 20% [24] in the autumn and winter months, on account of climate change. With an extensive road network of 5306 km comprising dual carriageways, motorways, and single lane roads [25], an increase in high intensity stormflow conditions could further exacerbate the volumes of road runoff and contaminants entering waterways.

Sustainable drainage systems (SuDS), Green Infrastructure (GI) and Low Impact Development (LID) are among available approaches to managing stormwater runoff in a way that minimises environmental impact. SuDS technologies such as constructed wetlands, permeable pavements and filter drains are designed to manage both quality and quantity of urban surface runoff [26]. While installation of these technologies in new developments is feasible and required in Ireland [27], retrofitting them into existing infrastructure can be challenging and resource intensive. A new material called BlueLay was evaluated as a potential retrofit SuDS material. It is a highly porous medium made of compressed rockwool and held between sheets of glass fibre.

In this paper, findings are presented from tests where BlueLay is dosed with TSS and a subset of heavy metals (Cu, Zn, Pb, Cd). Initially, the hydraulic performance of the BlueLay was tested in terms of infiltration volume, absorption capacity and runoff attenuation compared with asphalt (Stone Mastic Asphalt - SMA 10) slabs of the same area. Results of hydraulic tests on BlueLay indicated that attenuation of water was higher (20–27%) in short duration storms (5 min). High intensity (172 mm/h) short duration (5 min) storm conditions were therefore adopted for this study to evaluate the water quality performance of BlueLay, compared to asphalt. The following research questions were investigated:

1. How does dry BlueLay compare with asphalt roads in terms of:
   (a) Total Suspended solids removal?
   (b) Heavy Metals removal?
2. Does pollutant removal efficiency (BlueLay) vary when saturated with water?
3. Does pollutant removal efficiency (BlueLay) vary when saturated with pollutants?

2 Materials and Methods

2.1 Pollutant Test Rig—Rainfall Simulator

Rainfall simulators (Fig. 1) were developed in-house to generate short duration high intensity rainfall (179 mm/h or 0.2 L/min). Each rainfall simulator consisted of an acrylic tube 0.5 m in length with a diameter of 100 mm fitted with 3-D printed nozzles and base plate. An inflow and overflow pipe ensure that a constant head was maintained for each rainfall event. All plumbing fixtures were made from plastic/acrylic to remove any possible metal contamination. Each simulator was supplied with deionised (DI) water from a header tank with an on-off valve arrangement. DI water was used to ensure consistency and repeatability of experiments.

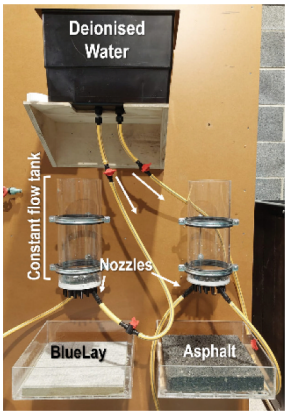

Fig. 1. Constant head rainfall simulators testing BlueLay (left) and Asphalt (right) slabs (305 mm × 305 mm).

Each test slab (BlueLay/Asphalt) was placed below the rainfall simulators in an acrylic tray with drain holes for effluent collection. Slabs were elevated to a slope (0.5%) to emulate the minimum slope of a road surface in Ireland [28]. Pollutants were dosed on the surface of test slabs. To ensure there were no complex interactions between the different pollutant classes, TSS and metals were dosed separately. BlueLay and asphalt were tested under three different scenarios to simulate real life rainfall conditions (Fig. 2). Runoff and infiltration effluents were then analysed (Sect. 2.3) for TSS and metals to assess pollutant removal efficiency of each slab.

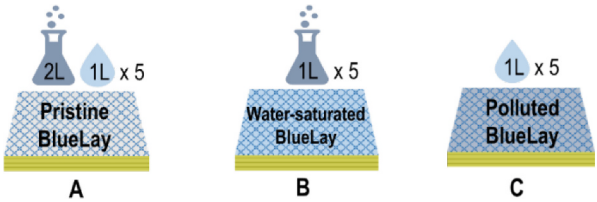

Fig. 2. Test scenarios for BlueLay (Asphalt tested only for scenario A). (A) Pristine material subjected to 2 L pollutant (flush) followed by 5 rain events. (B) Material after high rainfall period subjected to 5 pollutant (flush) events. (C) Material after high rainfall and dry periods subjected to 5 rain events

2.2 Synthetic Runoff—TSS and Heavy Metals

DI water was used while making synthetic runoff (pollutant) solutions for both TSS and metals solutions to ensure repeatability and consistency. The ranges of TSS and dissolved metals typically found in road runoff are provided in Table 1.

Table 1. Ranges of heavy metals and TSS found in road runoff based on literature

Pollutant	Units	Runoff range [16, 26]	Legislative requirements (S.I. 272/2009)
Cd	µg/L	<4.8	<0.8
Cu	µg/L	<136	<5
Pb	µg/L	<179	<20
Zn	µg/L	34–497	<30
TSS	mg/L	75–774	<30*

*Guideline recommends TSS <25 mg/L for freshwater bodies, <30 mg/L for coastal and transitional waters.

Kaolin ($Al_2Si_2O_5(OH)_4$), a naturally occurring hydrous aluminium silicate mineral, was used to create synthetic runoff for TSS tests. Kaolin has a particle size of <5 µm. BlueLay was tested under all three scenarios for TSS removal efficiency. Asphalt was not tested for scenarios B and C as saturation was not possible.

A trace metal solution was prepared with metal standards of Zn, Cu, Cd, and Pb. Stock solutions for each heavy metal were prepared separately using 1000 mg ampoules (Merck) of dissolved metal solutions. Pollutant mixes were made with equal volumes of each metal type with concentrations based on literature review (Table 1). Tests were carried out in triplicate to ensure repeatability and reproducibility. Pollutant mixes for TSS and heavy metals were made in 3 concentrations: C1, C2, and C3 (Table 2). C3 was the highest concentration factoring in 20% increase for climate change [24].

Table 2. Concentrations of synthetic pollutant mixes for testing

Heavy metals (μg/L)	C1	C2	C3
Cd	1	5	10
Cu	50	110	170
Pb	50	100	145
Zn	30	315	600
TSS — Kaolin (mg/L)	75	350	800

2.3 Analytical Methods

TSS was measured using EPA method 160.2, a gravimetric method for TSS analysis (LOD = 0.1 mg/L). Samples were filtered through 0.05 μm Millipore filter paper. Since the presence of TSS in a sample affects its light scattering properties, the amount of light scatter is proportional to the concentration of suspended solids in the sample. Therefore, turbidity of samples was also measured in addition to TSS.

To calibrate a UV-Vis spectrophotometer (Hach DR 2800) for metals analyses, a preliminary analysis was done by ICP-MS (Thermo Scientific iCAP-Q ICP-MS) at the National Centre for Isotope Geochemistry in a high collusion matrix (cell collision mode). The analytical methods and limits of detection (LOD) for Hach DR 2800 are provided in Table 3. Effluent samples from each test were filtered (0.45 μm cellulose filter) and acidified prior to analysis (2.5 M Nitric acid).

Table 3. Analysis methods and LODs for metals using UV — Vis spectrophotometer

Metal	Analysis methods	Limits of detection (LOD)
Cd	Dithizone method (8017)	0.7–80.0 μg/L
Cu	Porphyrin method (8143)	1–210.0 μg/L
Pb	LeadTrak method (8317)	5–150 μg/L
Zn	Zincon method (8009)	0.02–3.0 mg/L

3 Results and Key Findings

3.1 TSS Removal

Scenario A: Pristine dry material (Fig. 3a, b). Dry pristine BlueLay (Fig. 3a) achieved 89% removal for the highest concentration (800 mg/L) of applied TSS after one rain event and an overall of 85% TSS removal after five rain events. For the same concentration, asphalt (Fig. 3b) removed 15% TSS after the first rain event and an overall of 7%

after the fifth rain event. For the lowest concentration (≤75 mg/L), BlueLay removed 84% and 71% TSS after the first and fifth rain events respectively. This shows that BlueLay attenuated low concentrations (≤75 mg/L) of TSS to within WFD guidelines (<30 mg/L). For medium (350 mg/L) and high (800 mg/L) concentrations, dry BlueLay exceeded guidelines by ~26 mg/L and ~ 60 mg/L (first rain event). In comparision, dry asphalt slabs exceeded guidelines by ~268 mg/L and ~648 mg/L.

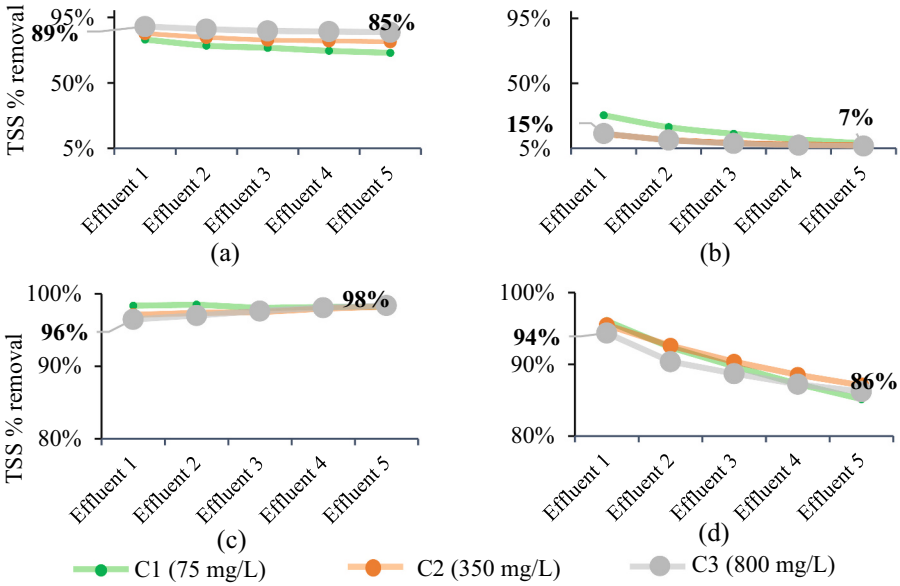

Fig. 3. TSS % removal by (**a**) dry pristine BlueLay, (**b**) dry asphalt, (**c**) water-saturated BlueLay and (**d**) pollutant-saturated BlueLay when subjected to three different TSS concentrations followed by five high intensity short duration rain events. Results are shown for all rain events.

Scenario B: Water-saturated BlueLay (Fig. 3c). Water-saturated BlueLay achieved 97 ± 1% removal after first flush and an overall of 98% removal after the fifth flush event for all applied concentrations. BlueLay attenuated TSS concentrations (≤ 800 mg/L) to <30 mg/L even after five flush events.

Scenario C: Polluted BlueLay (Fig. 3d). Polluted BlueLay attenuated TSS for medium and low concentrations to within WFD guidelines (<30 mg/L). For high concentrations, BlueLay removed 94% TSS after the first rainfall event an overall of 86% TSS after the fifth rainfall event, but the TSS in the effluent exceeded WFD guidelines by ~15 mg/L.

3.2 Heavy Metals Removal

BlueLay in its dry state attenuated all four metals (Cu, Zn, Cd, Pb) to values well within the legislative [20] limits (Table 1) for medium (C2: 520 μg/L) and low range (C1: 132 μg/L) metals concentrations (Table 2).

When high concentration (C3: 925 μg/L) metals solution was applied to BlueLay, after the first rain event effluents (Fig. 4a) showed Cu, Zn and Cd concentration exceeding legislative requirements by 1.4 μg/L (Cu), 9.2 μg/L (Zn) and 0.4 μg/L (Cd). Pb was removed to well within legislative limits (<20 μg/L).

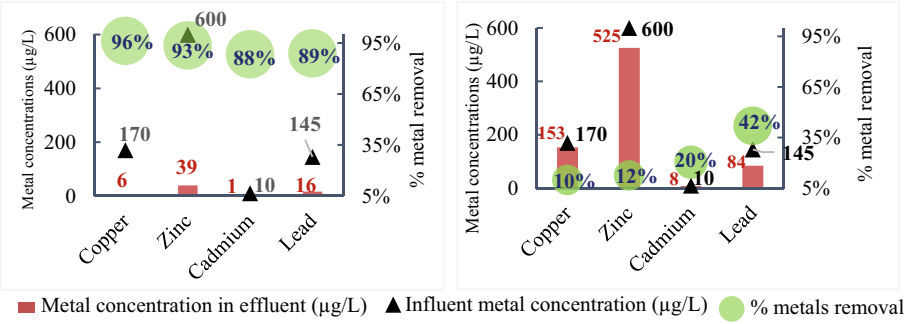

Fig. 4. Attenuation of high concentration metals (925 μg/L) by **(a)** dry BlueLay and **(b)** dry Asphalt after first rain event.

However, these values are considerably lower than the values obtained from the runoff from asphalt slabs of the same size, which exceeded the required limits by ~148.2 μg/L (Cu), ~495 μg/L (Zn), 7.17 μg/L (Cd) and 64 μg/L (Pb). Results shown in Fig. 4 are for a subset of Scenario A where metals concentrations in the effluents from BlueLay (Fig. 4a) and Asphalt (Fig. 4b) exceeded legislative limits.

4 Conclusions and Future Research

This study investigated the pollutant (TSS and heavy metals) removal performance of a porous medium, BlueLay to evaluate its potential as a sustainable drainage material. Attenuation values obtained in this study are for a slab of surface area 0.09 m^2 subjected to high intensity short duration rainfall (179 mm/h, 5 min).

Dry BlueLay attenuated influent TSS ≤ 75 mg/L to <30 mg/L (WFD guidelines). For medium (350 mg/L) and high (800 mg/L) concentrations, both dry BlueLay and asphalt exceeded guidelines. Exceedances from asphalt were ~ 10× and ~ 11× those from BlueLay. Partially or fully water saturated BlueLay attenuated TSS to <30 mg/L for influent concentrations ≤800 mg/L. When saturated with pollutant, effluent TSS concentrations exceeded guidelines by ~15 mg/L. Dry BlueLay effectively removed up to 98% of metal ions when subjected to metal concentrations ≤520 μg/L. For higher concentrations (C3: 925 μg/L), metal concentrations in effluents from BlueLay and asphalt exceeded legislative limits. Exceedances from asphalt were ~ 100× (Cu), 50×(Zn) and 20×(Cd) larger than those from BlueLay.

To meet regulatory requirements for higher concentrations of pollutants, the performance of BlueLay needs be complemented. From this perspective and for its potential application as a retrofit SuDS material, BlueLay is currently being tested for its performance when interfaced with other porous media such as soil, vegetation and gravel that

may be present next to road surfaces. Soil is known to attenuate pollutants like metals by adsorption [29]. Gravel and vegetation can promote sedimentation of TSS by reducing water velocity [30]. It remains to be seen whether these properties could improve the performance of BlueLay.

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